Multi-Hub Shield

DEVLAB FACTOR

Mfr. Part #: UE0064

Product Reference Manual (V1.0)

Description

This is a modular shield expansion board designed to be compatible with various microcontroller platforms, including Pulsar, Xiao, Pi Pico, Feather, Devkit V4, and Mikro Bus. The shield provides a wide range of interfaces and features for prototyping and development, making it an ideal choice for makers, educators, and developers looking to expand their projects with additional connectivity and functionality.

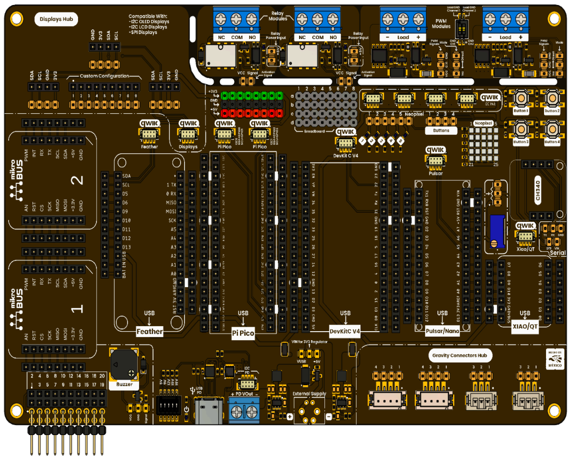
The board is intended for makers, educators, and developers who require a flexible and scalable solution to expand connectivity, control, and interaction capabilities within embedded applications.

DevLab format compatibility.

Simplicity and compatibility are the primary objectives of the DevLab form factor. It offers a board layout that is compact and optimized for serial communication, which includes I²C and SPI interfaces.

This format enables the establishment of rapid and dependable connections via a Protocol I²C connector that is entirely compliant with the Qwiic and STEMMA standards.

This design guarantees efficient prototyping, accessibility, and simplicity of integration across a variety of devices and modules.



Applications

- Prototyping: Quickly develop and test ideas.
- Education: Suitable for learning microcontroller basics.
- Wearables: Compact and versatile for wearable devices.
- Displays: Use the LED matrix for simple visual output.

Software Support

- The board is supported by both official and community-developed resources, including libraries, examples, and integration guides for common development environments such as Arduino and PlatformIO.

Hardware Features

- **Support Shield Format:** Pulsar, NANO, Xiao, QT, Pi Pico, Feather, Devkit V4, Mikro Bus
- **Connectivity:** I2C, SPI, UART, GPIO, ADC
- **LED Matrix:** 5×5 RGB LED matrix for visual feedback
- **DevLab Ecosystem:** 7 compatible connectors (JST 1 mm) with QWIIC–STEMMA compatibility
- **Relay Modules:** 2 relay modules
- **Display Hub:** I2C 128×64 OLED display, display customization configuration, TFT support
- **PWM Interface:** 2 PWM modules
- **Potentiometer:** 1 potentiometer for analog input
- **Gravity Interface:** 4 Gravity connectors for sensor integration (2×4 and 2×3 connectors)
- **PWM Buzzer:** 1 PWM buzzer for audio feedback
- **Expansion Lines:** 4 breadboard lines for custom circuit connections
- **Expansion Connectors:** 1 connector with 20 pins for additional modules and shields
- **Buttons:** 4 buttons for user input and control
- **LEDs:** 4 LEDs for status indication and feedback
- **Power Delivery:** Supports USB-C, 3.3 V, 5 V, and external power supply options for versatile power management

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1.1 Accessories

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- 1x QWIIC 4-pin JST 1.0 mm Harness

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2.1 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
VIN_PD	USB-C PD Input Voltage	5	5 / 9 / 12 / 15 / 20	20	V
VIN_EXT	External Supply Input Voltage	5.0	5.0	5.5	V
VOUT_5V	Regulated 5V Output Rail	4.85	5.0	5.15	V
VOUT_3V3	Regulated 3.3V Output Rail	3.20	3.30	3.40	V
IOUT_5V	Recommended 5V Rail Output Current	1.2	1.6	2.0	A
IOUT_3V3	Recommended 3.3V Rail Output Current	1.2	1.6	2.0	A
TA	Operating Ambient Temperature	0	25	70	°C
VIH	Digital Input High Level	0.7 × VDD	—	VDD	V
VIL	Digital Input Low Level	0	—	0.3 × VDD	V
FI2C	I²C Bus Frequency	100	400	1000	kHz
VQWIIC	QWIIC/STEMMA Connector Supply Voltage	3.3	3.3 / 5.0	5.0	V
VRELAY	Relay Coil Supply Voltage	4.75	5.0	5.25	V
IRELAY	Relay Coil Current (per channel)	—	21.1	—	mA
VLOAD_RELAY	Relay Switched Load Voltage	—	—	30 (DC) / 125 (AC)	V
ILOAD_RELAY	Relay Switched Load Current	—	—	1 @ 30VDC / 0.5 @ 125VAC	A
VPWM	PWM Output Load Voltage	0	VDD	30	V

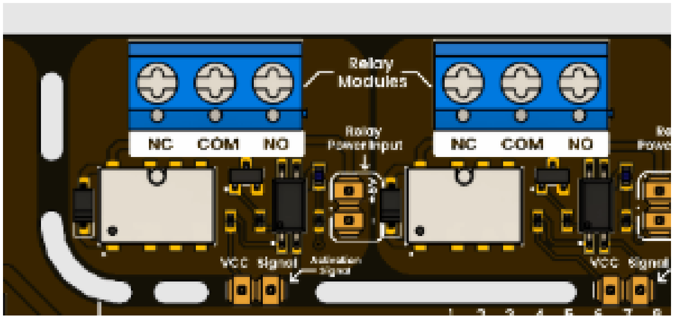
IPWM	PWM Load Current	—	Application-defined	10	A
VLED_MATRIX	WS2812B LED Matrix Supply Voltage	4.5	5.0	5.5	V
ILED_MATRIX	WS2812B Matrix Current (5×5 Full White)	—	Application-dependent	200	mA
VLED_IO	WS2812B Data Input Logic Level	—	3.3 / 5.0	VDD	V
ILED_IND	Status LED Current	—	2–10	—	mA
VBUZZ	PWM Buzzer Operating Voltage	3.0	3.3 / 5.0	5.0	V
FBUZZ	PWM Buzzer Drive Frequency	0.5	-	500	kHz
VBTN	Push Button Logic Voltage	0	VDD	VDD	V
VADC	ADC Input Voltage Range	0	—	VDD_ADC	V
RVR	Trimpot Resistance	—	10	—	kΩ

3. Functional Overview

This shield is designed as a development-focused platform that integrates multiple functional blocks into a single hardware solution. Its architecture supports rapid prototyping, educational applications, and system validation by combining control, visualization, interaction, and expansion capabilities. The board consolidates essential features such as relay control, display interfaces, signal modulation, and modular connectivity, enabling users to build complete embedded systems without requiring extensive external hardware.

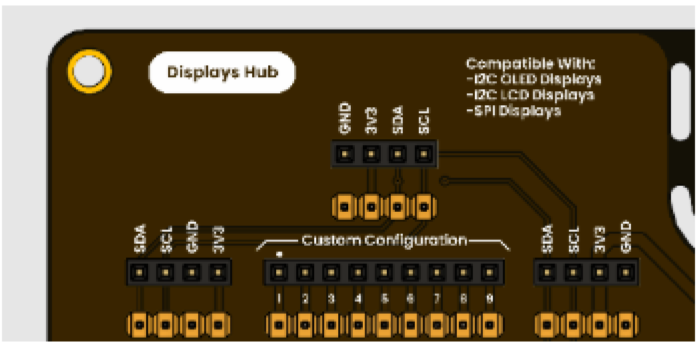
3.1 Relay Control Section

The board integrates two relay channels designed for switching external loads through isolated control signals. These relays allow the microcontroller to safely interface with higher-power devices by separating the control domain from the load domain. Each channel provides standard switching terminals, enabling flexible connection to external systems such as lighting, actuators, or low-power AC/DC equipment. This section is positioned to maintain clear separation between signal routing and load paths, improving reliability during operation.



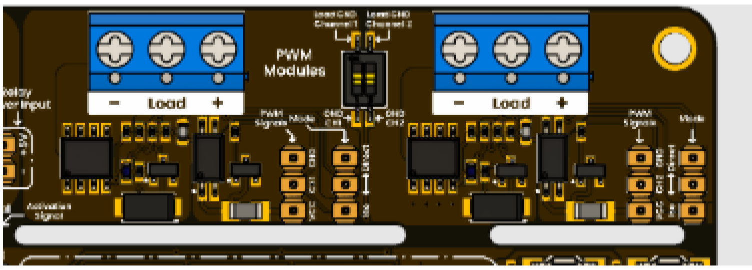
3.2 Display and Visualization Section

A dedicated display interface area is provided to support visual output devices, including I²C-based OLED displays and compatible TFT modules. The layout is optimized to simplify integration and minimize signal complexity, enabling stable communication for real-time data visualization. This section is intended for implementing user interfaces, diagnostics, and monitoring systems within embedded applications.



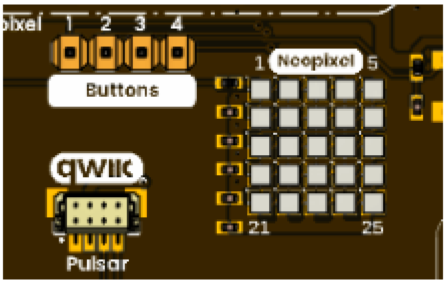
3.3 PWM and Signal Control Section

The board includes PWM-capable outputs intended for variable signal control. These outputs enable applications such as brightness regulation, proportional control, and signal-based interfacing with external driver stages. The routing ensures accessibility and compatibility with standard control requirements, allowing integration with a wide range of peripherals that rely on pulse-width modulation.



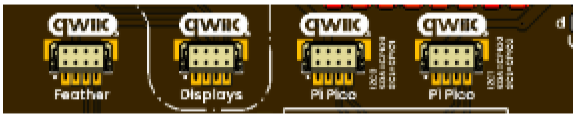
3.4 User Interaction and Feedback Section

Integrated interaction elements include a 5×5 RGB LED matrix, multiple push buttons, and status indicator LEDs. These components provide a direct interface for user input and system feedback without requiring additional hardware. The arrangement supports intuitive interaction, making the board suitable for demonstration systems, user interfaces, and rapid validation of embedded behaviors.



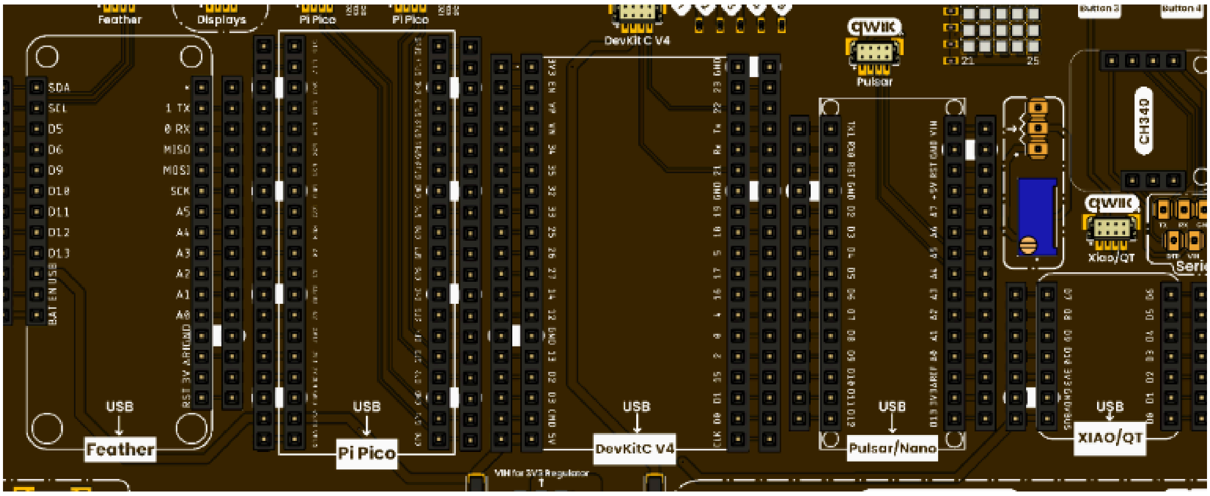
3.5 DevLab / QWIIC Connectivity Section

The board incorporates multiple JST 1 mm connectors aligned with QWIIC and STEMMA QT standards, enabling simplified I²C-based expansion. These connectors support plug-and-play integration of compatible modules such as sensors and displays, reducing the need for manual wiring. The distributed placement across the board facilitates modular design and efficient system scaling.



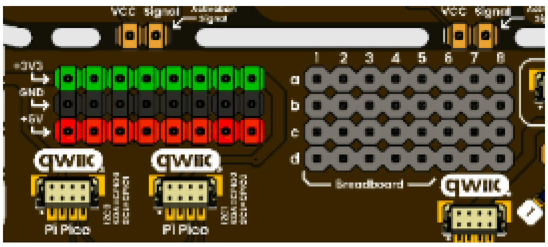
3.6 Platform Compatibility Section

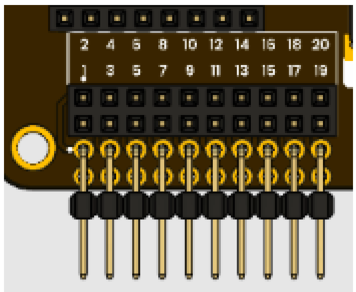
The central layout is organized into multiple compatibility zones to support a variety of development platforms, including Feather, Pi Pico, Devkit V4, Pulsar/Nano, XIAO/QT, and MikroBus formats. Each zone exposes the required pin mappings for proper integration, allowing the shield to function as a universal interface across different ecosystems. The silkscreen serves as a reference for correct placement and alignment.



3.7 Expansion and Prototyping Section

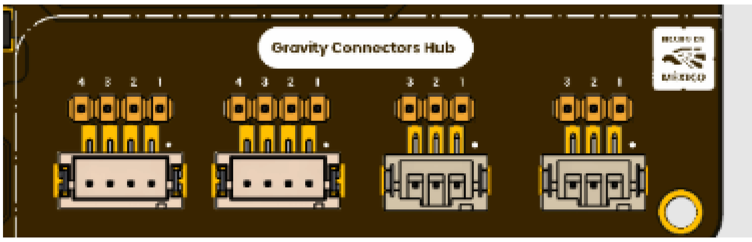
A prototyping area is included to support custom circuit development directly on the board. This section provides access to power rails and signal points, enabling quick integration of additional components without external prototyping tools. It is designed to complement the modular nature of the board, allowing flexible experimentation and system extension.





3.8 Sensor Interface Section

Gravity-style connectors are integrated to support the rapid connection of sensors and peripheral modules. These connectors simplify wiring and provide standardized access to power and signal lines, making them suitable for data acquisition and educational applications where multiple sensors are used simultaneously.



3.9 Power Distribution Section

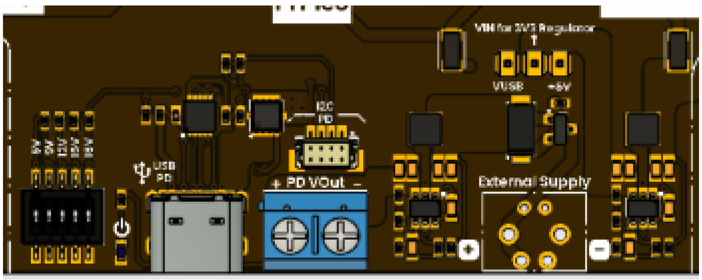
The board integrates a flexible power distribution architecture designed to support operation from both USB-C Power Delivery (PD) and external power sources. The system distributes energy across independent 5 V and 3.3 V domains, allowing compatibility with mixed-voltage peripherals, embedded platforms, and expansion modules.

The power subsystem includes dedicated voltage regulators, protection circuitry, source-selection logic, and configurable routing headers. This architecture allows the board to operate under different supply conditions while maintaining stable voltage regulation and controlled power distribution across the system.

The design supports three primary operating modes:

- USB-C Power Delivery operation
- External supply operation
- Simultaneous PD and external supply operation with automatic source priority

The board also integrates MOSFET-based protection circuitry to prevent reverse current conditions and undesired interaction between power sources.



3.9.1 USB-C Power Delivery Section

The USB-C power stage is based on the HUSB238 USB Power Delivery controller. This device negotiates voltage profiles directly from compatible USB-C PD chargers and exposes the negotiated voltage to the main VIN distribution rail.

Unlike fixed USB 5 V systems, the HUSB238 allows dynamic voltage selection through a resistor configuration network connected to the configuration pins of the controller. The selected voltage depends on the installed resistor or selector configuration.

Supported Power Delivery profiles include:

PD Profile	Output Voltage
5 V	Selectable
9 V	Selectable
12 V	Selectable
15 V	Selectable
18 V	Selectable
20 V	Selectable

The PD voltage is not fixed by default and depends entirely on the selected configuration and charger capabilities. The negotiated voltage is routed to the VIN rail and can subsequently feed the onboard regulators to generate regulated system voltages.

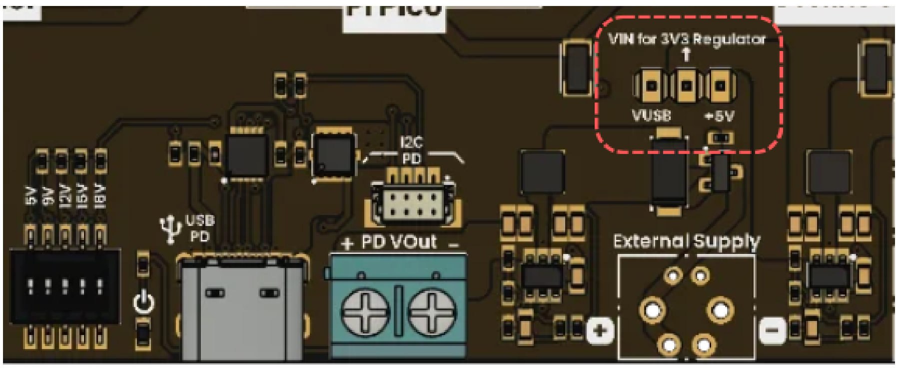
Recommended Power Supply Configuration

For standard operation using USB-C PD:

- Configure jumper: **VUSB → VREG**

This configuration routes the negotiated PD voltage through the onboard regulator stages, enabling stable generation of:

- Regulated 5 V rail
- Regulated 3.3 V rail



This mode is recommended for general embedded applications and mixed-voltage peripheral systems.

3.9.2 Voltage Regulation Stage

The board integrates two independent TPS54302DDCT synchronous buck regulators to generate regulated voltage domains from the VIN distribution rail.

The regulation subsystem is divided into:

Rail	Purpose
5 V	Peripheral and expansion power
3.3 V	Logic and low-voltage devices

Each regulator stage includes:

- Input filtering capacitors
- Bootstrap capacitor
- Feedback compensation network
- Inductor output filtering
- Output bulk capacitors

This architecture provides stable operation under varying input voltages and dynamic load conditions.

5 V Regulation Stage

The first TPS54302 regulator generates the regulated 5 V rail used for peripheral distribution and external interfaces.

Features include:

- Wide input voltage compatibility
- High-efficiency synchronous conversion
- Stable regulation under PD input conditions

The regulator output is distributed to:

- Expansion headers
- Peripheral connectors
- System 5 V rail
- Secondary regulation stages

3.3 V Regulation Stage

The second TPS54302 regulator generates the regulated 3.3 V rail used for logic-level devices and low-voltage peripherals.

This rail powers:

- Microcontroller logic
- I2C peripherals
- Sensors
- Communication interfaces

- Low-voltage expansion modules

The regulation stage is isolated from the main VIN rail through controlled routing and filtering to improve stability and noise performance.

3.9.3 External Supply Section

The board also supports direct external power injection through the terminal block interface. This mode allows operation without requiring a USB-C PD source.

The external supply input is designed primarily for regulated 5 V operation.

When operating from the external supply input, the onboard regulation stages can still generate the required 3.3 V rail depending on the selected jumper configuration.

Recommended External Supply Configuration

For external 5 V operation:

- Configure jumper: **5V → 3V3 regulator enabled**

In this configuration:

- The external 5 V rail powers the system directly
- The onboard regulator generates the 3.3 V logic rail
- Peripheral distribution remains active

This configuration is recommended for standalone systems or applications powered from laboratory supplies, batteries, or industrial power systems.

3.9.4 Source Selection and Protection Circuitry

The board integrates automatic source-selection circuitry to manage coexistence between USB-C PD and external power sources.

The source-selection stage uses an AO3407 MOSFET array combined with passive protection circuitry to implement the following:

- Reverse current protection
- Source isolation
- Automatic voltage path selection
- Controlled current flow
- Priority management

The MOSFET stage prevents undesired backfeeding between power sources and protects the regulators during transitions between supply modes.

Power Path Operation

When a USB-C PD source is connected:

- The PD source becomes the primary power source
- The AO3407 stage isolates the external supply path
- System power is routed from the negotiated PD rail

When the PD source is disconnected:

- The external supply path automatically becomes active
- Power transitions without manual intervention
- System operation continues from the external source

This architecture allows uninterrupted operation and improves system reliability under mixed-source conditions.

3.9.5 Power Configuration Cases

Case 1. USB-C Power Delivery Operation

This mode uses USB-C PD as the primary power source.

Recommended Configuration

VUSB → VREG

Operation

- HUSB238 negotiates the configured PD voltage
- VIN rail receives negotiated voltage
- Regulators generate stable 5 V and 3.3 V rails
- Recommended for standard operation

This mode provides maximum flexibility and allows operation from high-voltage USB-C PD chargers.

Case 2. External 5 V Supply Operation

This mode uses the external terminal block as the primary power source.

Recommended Configuration

5V → 3V3 regulator enabled

Operation

- External 5 V source powers the system
- 3.3 V rail generated locally by regulator
- Suitable for battery systems and laboratory supplies

This configuration allows operation without requiring USB-C PD negotiation.

Case 3. USB-C PD + External Supply

This mode allows simultaneous connection of both power sources.

Operation

- USB-C PD source receives priority
- External supply acts as backup source
- Automatic switching occurs if PD power is removed
- MOSFET stage prevents reverse current flow

This configuration is useful in redundant or failover power applications where uninterrupted operation is required.

3.9.6 Power Distribution Considerations

To ensure stable operation, the following considerations are recommended:

- Verify jumper configuration before applying power
- Use regulated external supplies (only +5V)
- Avoid exceeding regulator thermal limits
- Ensure PD chargers support the selected voltage profile
- Use appropriate cable quality for high-current PD operation

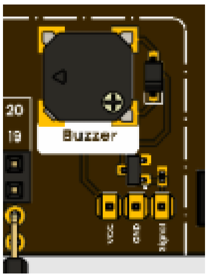
For high-power configurations:

- Ensure adequate cooling
- Verify connector current capability
- Avoid simultaneous excessive peripheral loading

The board architecture is designed to support flexible embedded development while maintaining safe operation across multiple voltage domains and power-source configurations.

3.10 Audio Feedback Section

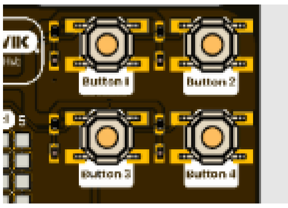
An onboard PWM-controlled buzzer provides audible feedback capabilities for system status, alerts, or interaction cues. It enables simple sound generation controlled directly by the microcontroller, supporting user-oriented applications and debugging scenarios.



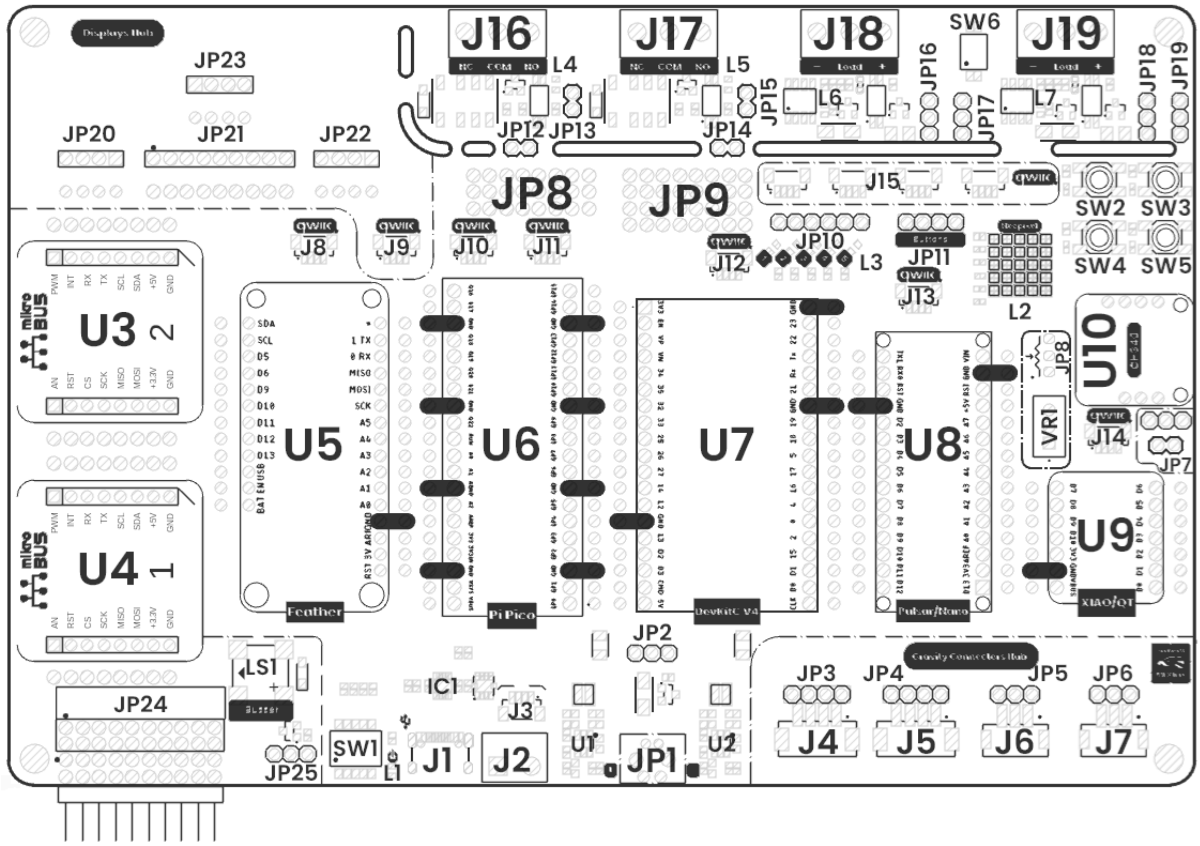
3.11 Button Interface Section

The board integrates four user buttons designed for direct interaction with the system. These buttons operate using a push-type mechanism and are electrically configured for digital input detection, allowing reliable state changes when pressed or released.

The implementation supports typical push–pull signal behavior through the microcontroller configuration, enabling straightforward integration for user control, menu navigation, or event triggering. This section is intended to provide immediate manual input without requiring additional external components, simplifying development and testing workflows.



3.13 Board Topology



Top View of Board Topology

Views of Topology

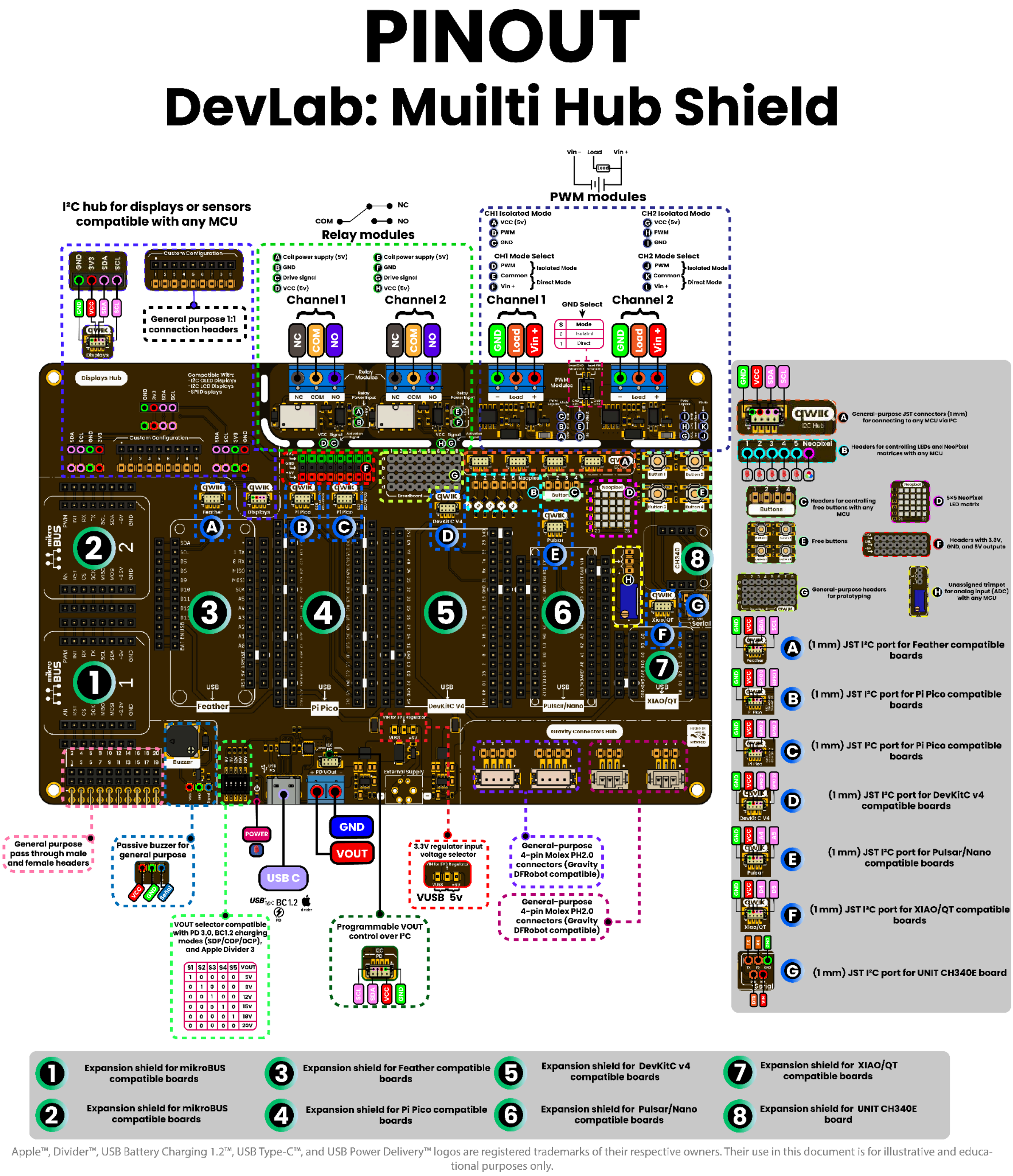
Table 3.2.1 - Components Overview

Ref.	Description
U3, U4	mikroBUS™ Shield
U5	Feather Shield
U6	Pi Pico Shield
U7	DevKit V4 Shield
U8	Pulsar Shield (NANO form factor)
U9	XIAO/QT Shield

U10	UNIT CH340E USB-to-Serial Shield
SW1	DIP switch for selecting PD VUSB output (5V, 9V, 12V, 15V, 20V)
J1	USB connector for Power Delivery (PD) input
J2	Terminal block for PD VUSB output (5V, 9V, 12V, 15V, 20V)
J3	QWIIC connector (JST 1.0 mm pitch) for I²C – HUSB238 PD controller
J4, J5, J6, J7, JP3, JP4, JP5, JP6	2.54 mm JST Gravity-compatible connector hub
J8, J9, J10, J11, J12, J13, J14	QWIIC connectors (JST 1.0 mm pitch) for I²C
J15	QWIIC parallel bus connector (JST 1.0 mm pitch) for I²C
JP9, JP24	Auxiliary 2.54 mm pins for general-purpose use
JP20, JP21, JP22, JP23	Auxiliary connectors for I²C or SPI LCDs/displays
SW2, SW3, SW4, SW5, JP11	General-purpose push buttons
JP10	Signal header for general-purpose LEDs and WS2812B 5x5 matrix (L2)
J16, J17	Terminal blocks for relay outputs
J18, J19	Terminal blocks for PWM module outputs
VR1, JP8	10k trimpot for ADC applications
LS1, JP25	SMD buzzer for general-purpose use
JP8	Power rail supply (3.3V, 5V, GND) for external devices

4 Connectors & Pinouts

4.1 General Pinout



Multi-Shield General Pinout

4.2 Pinout General Description

Label	Description	Typical Use
GND	Ground reference	Common return path for all signals
3V3	3.3V regulated output	Power for sensors, logic devices
VIN / 5V	Main input voltage	External power supply, USB input
SDA	I²C Data Line	Communication with sensors (EEPROM, IMU, etc.)
SCL	I²C Clock Line	Synchronization for I²C devices
TX	UART Transmit	Serial communication to PC or modules
RX	UART Receive	Serial data input
GPIO	General Purpose I/O	Digital input/output (buttons, LEDs)
ADC	Analog Input	Sensor reading (voltage, temperature, etc.)
PWM	PWM Output	Motor control, dimming LEDs, buzzer
INT	Interrupt Input	Event-driven signals from sensors
EN	Enable Control	Power enable/disable modules
RST	Reset Signal	Reset connected boards or modules
QWIIC SDA	I²C Data (JST 1mm)	Plug-and-play I²C modules
QWIIC SCL	I²C Clock (JST 1mm)	Plug-and-play I²C modules
NeoPixel DIN	Data input for LED	Control WS2812 LED strips/matrix
BTN1	Button Input	User interaction
BTN2	Button Input	User interaction
VOUT	Regulated output voltage	Power external circuits
USB VBUS	USB power line	Power from USB-C
LOAD+	Load positive terminal	External load connection
LOAD-	Load negative terminal	External load return

COM	Relay common	Switching circuits
NO	Relay normally open	Load activation
NC	Relay normally closed	Default load state
MOSI	SPI Master Out	Data to SPI devices
MISO	SPI Master In	Data from SPI devices
SCK	SPI Clock	SPI synchronization
CS	Chip Select	Select SPI device
BUZZ	Buzzer Output	Sound alerts, feedback
LED	Status LED	Visual indication
PROG	Programming Pin	Firmware flashing
ID	Board ID / Address	Device identification

5 Board Operation

5.1 Getting Started

This section provides a complete initial procedure for operating the Multi Hub Shield with a compatible microcontroller platform. The objective is to validate correct hardware integration, power distribution, and communication using a basic functional example.

The Multi Hub Shield is designed to operate with multiple development boards, including Pulsar C6 (ESP32-C6), ESP32 DevKit V4, XIAO, and other compatible formats. Each platform is supported through dedicated layout zones that expose the required pin mappings for proper operation.

The initial validation is centered on establishing I²C communication and verifying system behavior through a visual interface. For this purpose, the use of an I²C OLED display (128×64) is recommended, as it provides immediate feedback and simplifies debugging.

Hardware Requirements

- Multi Hub Shield
- Compatible development board (Pulsar C6, ESP32 DevKit V4, XIAO, or equivalent)
- I²C OLED display (128×64, SSD1306 recommended)
- USB-C cable for power and programming
- Optional jumper configuration or headers depending on platform

System Assembly

Begin by installing the selected development board into the corresponding compatibility zone on the shield. Ensure that all pins are correctly aligned and fully inserted. The silkscreen labeling on the board serves as the primary reference for positioning.

If required, configure jumpers or headers to enable communication lines such as I²C. Some platforms may require explicit routing of SDA and SCL signals depending on the internal pin configuration.

Once the microcontroller is installed, connect the OLED display to one of the DevLab/QWIIC connectors. These connectors provide direct access to the I²C bus and power lines, allowing a clean and solderless connection.

Electrical Connections

The QWIIC/DevLab connector provides the following signals:

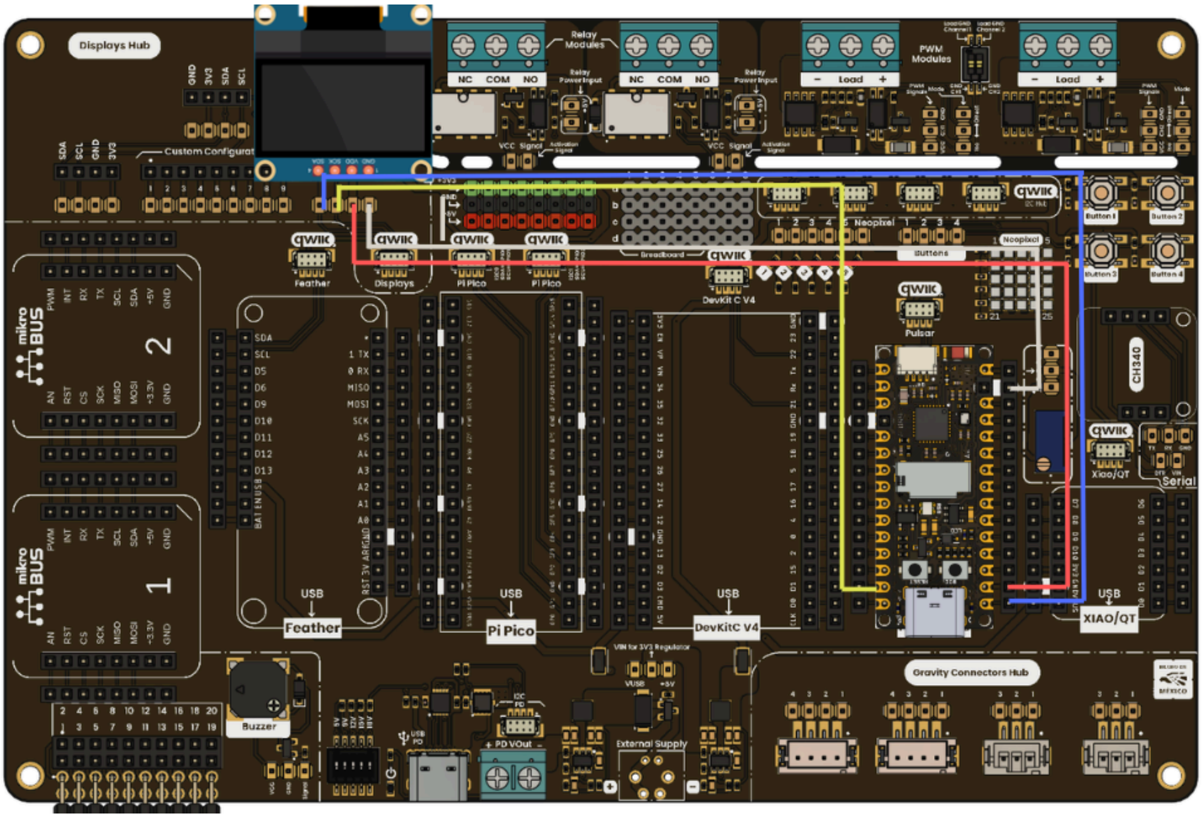
- VCC (3.3 V or 5 V depending on board configuration)
- GND
- SDA (data line)
- SCL (clock line)

For the Pulsar C6, the default I²C configuration is:

- SDA → GPIO6
- SCL → GPIO7

Ensure that the display module operates within the same voltage domain as the board. Incorrect voltage selection may prevent operation or damage the device.

Connection Diagram



Development Environment Setup

Open the Arduino IDE and configure the following:

- Board: ESP32C6 Dev Module
- Port: Corresponding serial port
- Libraries Required:
  - Wire (included by default)
  - Adafruit SSD1306
  - Adafruit GFX

Install the required libraries using the Arduino Library Manager if they are not already available.

Initial Test Application

The following example initializes the I²C interface and displays a basic message on the OLED. This confirms correct communication between the microcontroller and the display.

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define SDA_PIN 6
#define SCL_PIN 7

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

void setup() {
 Serial.begin(115200);

 // Initialize I2C
 Wire.begin(SDA_PIN, SCL_PIN);

 // Initialize display
 if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
  Serial.println("OLED initialization failed");
  while (true);
 }

 display.clearDisplay();
 display.setTextSize(1);
 display.setTextColor(SSD1306_WHITE);

 display.setCursor(0, 10);
 display.println("Multi Hub Shield");
 display.println("Initialization OK");
 display.display();
}

void loop() {
 // No repeated actions required
}
```

Code Description

The program begins by initializing the serial interface for debugging purposes. The I²C bus is then configured using the defined SDA and SCL pins corresponding to the selected platform.

The OLED display is initialized using its default I²C address (0x3C). If the display is not detected, the program halts execution, indicating a hardware or connection issue.

Once initialized, the display buffer is cleared, and a simple text message is rendered. This confirms that communication is established and that the system is functioning correctly.

Validation and Expected Behavior

After uploading the code:

- The OLED display should power on and show the initialization message.
- The serial monitor should not report errors if the device is correctly connected.
- If no display output is observed, verify wiring, I²C address, and voltage compatibility.

This validation step confirms:

- Correct power distribution
- Functional I²C communication
- Proper platform integration
- Working development environment

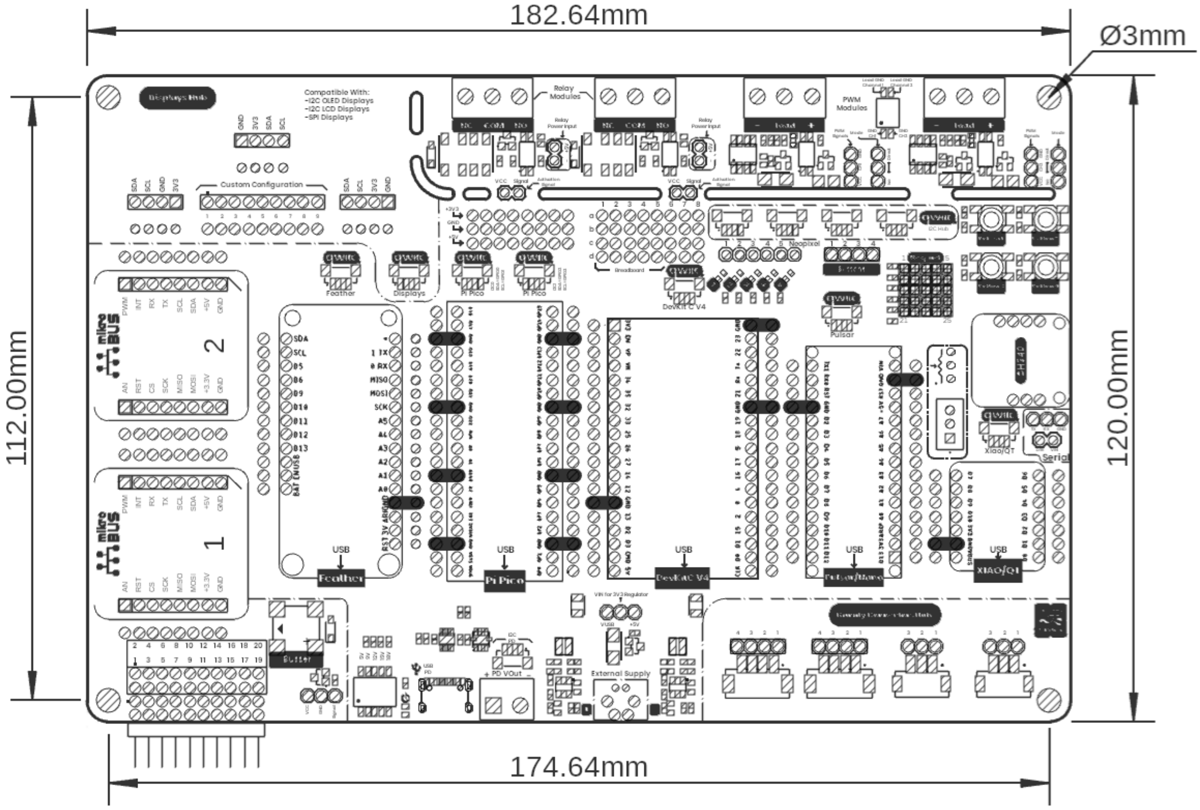
Next Steps

Once the initial test is successful, the system is ready for further development. Additional features of the Multi Hub Shield can be progressively enabled, including:

- Relay control for external loads
- PWM signal generation for control applications
- Sensor integration through DevLab or Gravity connectors
- User interaction using buttons and LED matrix

This structured approach ensures a reliable starting point before implementing more complex embedded applications.

6. Mechanical Information



Board Dimensions in Millimeters

Mechanical dimensions in millimeters

7 Company Information

Company name	UNIT Electronics
Company website	https://uelectronics.com/
Company Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX

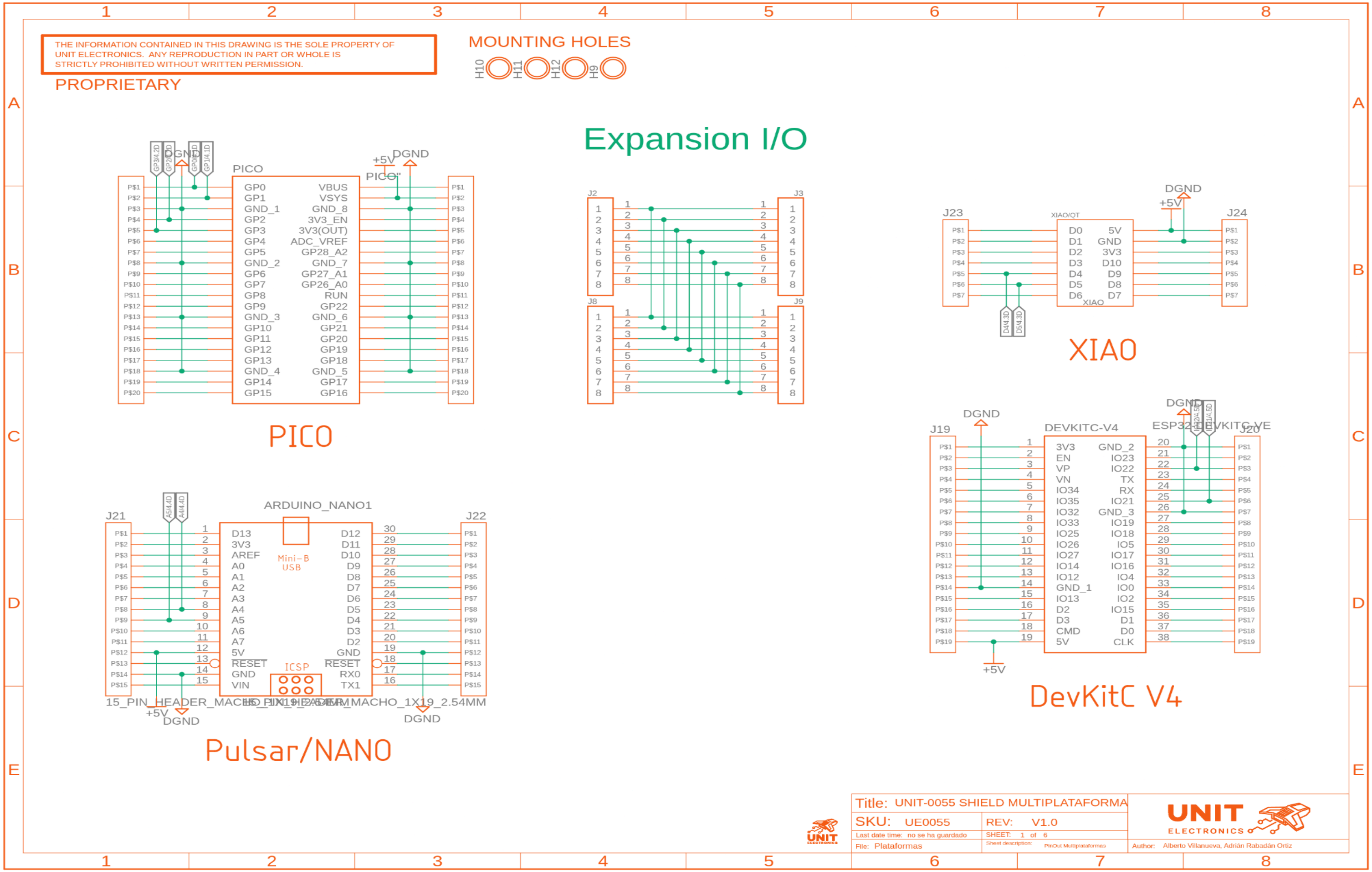
8 Reference Documentation

Ref	**Link**
Documentation	https://wiki.uelectronics.com/wiki/unit_devlab_multi_hub_shield
Github	https://github.com/UNIT-Electronics-MX/unit_devlab_multi_hub_shield/
Thonny IDE	https://thonny.org/
Arduino IDE	https://www.arduino.cc/en/software
Visual Studio Code	https://code.visualstudio.com/download

9 Appendix

9.1 Schematic

(https://github.com/UNIT-Electronics-MX/unit_devlab_multi_hub_shield/blob/main/hardware/unit_sch_v_3_2_0_ue0064_devlab_multihub_shield.pdf)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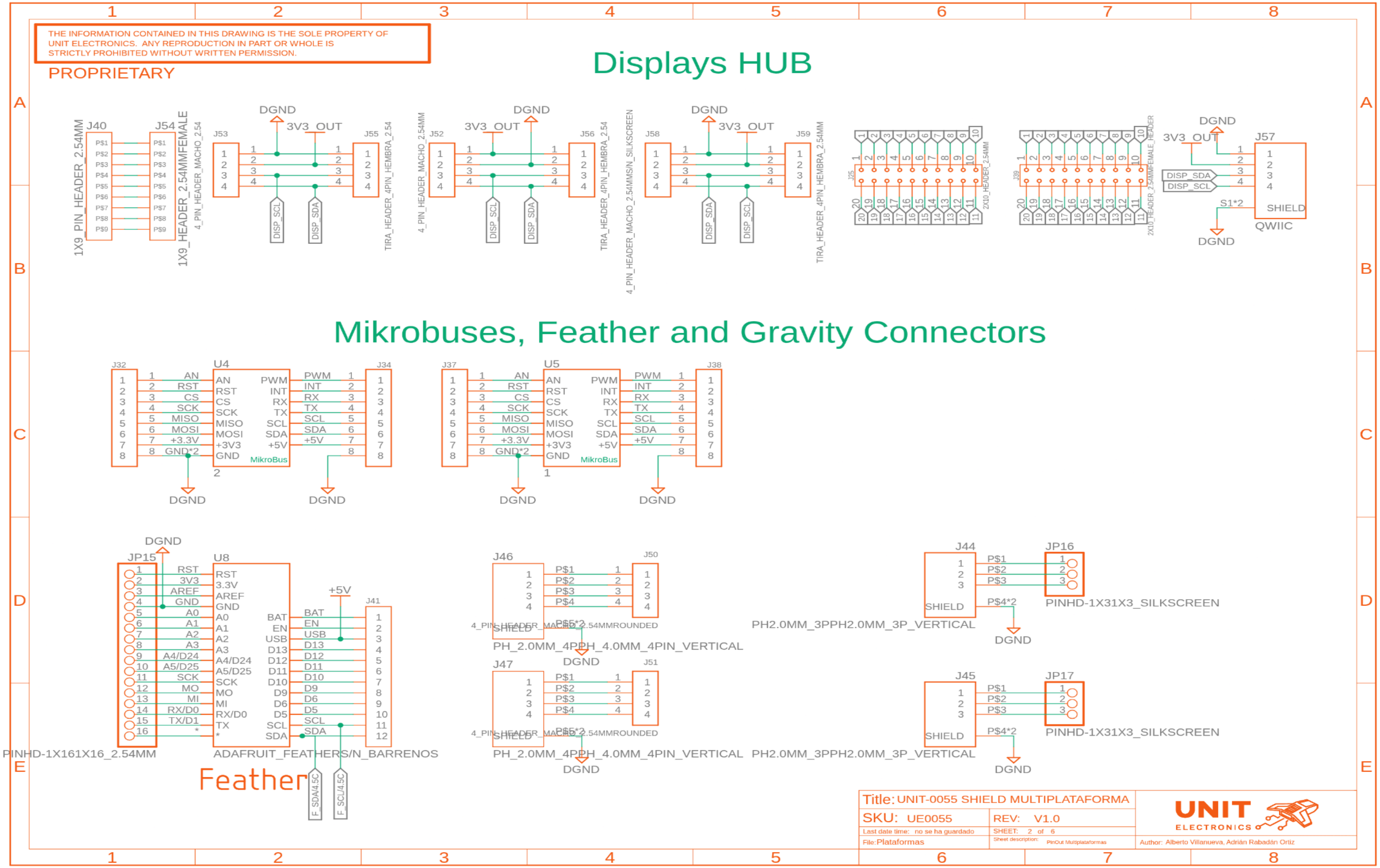
PROPRIETARY
MOUNTING HOLES
H10 H11 H12 H9
Expansion I/O
PICO
XIAO
DevKitC V4
Pulsar/NANO
ARDUINO_NANO1
DEVKITC-V4
J2 J3 J8 J9 J19 J20 J21 J22 J23 J24
DGND
+5V
Title: UNIT-0055 SHIELD MULTIPLATAFORMA
SKU: UE0055
REV: V1.0
SHEET: 1 of 6
File: Plataformas
Author: Alberto Villanueva, Adrián Rabadán Ortiz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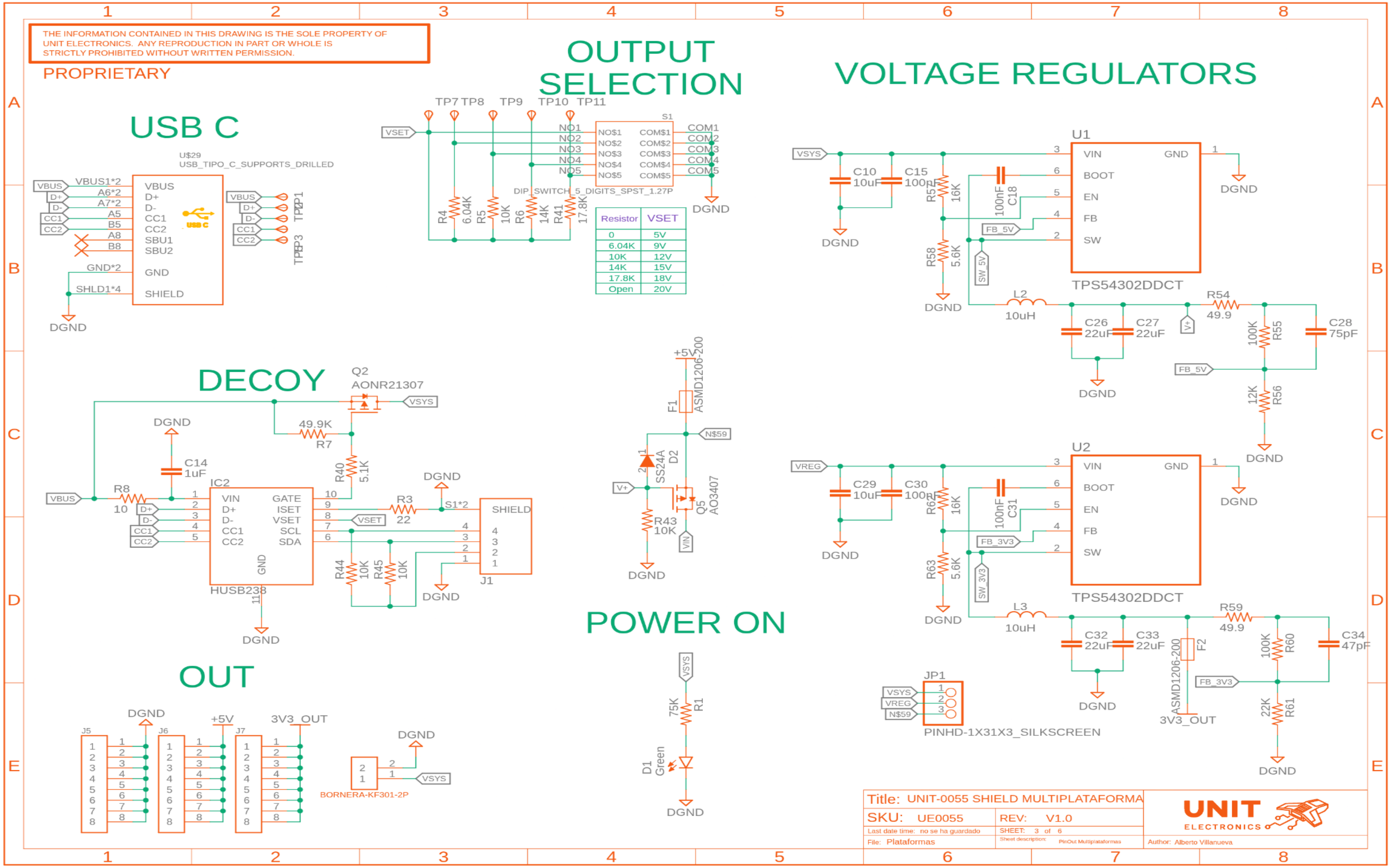
PROPRIETARY
Displays HUB
Mikrobuses, Feather and Gravity Connectors
Feather
DGND
3V3_OUT
DISP_SCL
DISP_SDA
J40
J54
J53
J55
J52
J56
J58
J59
J57
QWIIC
SHIELD
J32
U4
J34
J37
U5
J38
MikroBus
JP15
U8
J41
ADAFRUIT_FEATHERS/N_BARRENOS
PINHD-1X161X16_2.54MM
J46
J50
J47
J51
PH_2.0MM_4PPH_4.0MM_4PIN_VERTICAL
J44
JP16
J45
JP17
PINHD-1X31X3_SILKSCREEN
PH2.0MM_3PPH2.0MM_3P_VERTICAL
Title: UNIT-0055 SHIELD MULTIPLATAFORMA
SKU: UE0055
REV: V1.0
SHEET: 2 of 6
File: Plataformas
Author: Alberto Villanueva, Adrián Rabadán Ortiz



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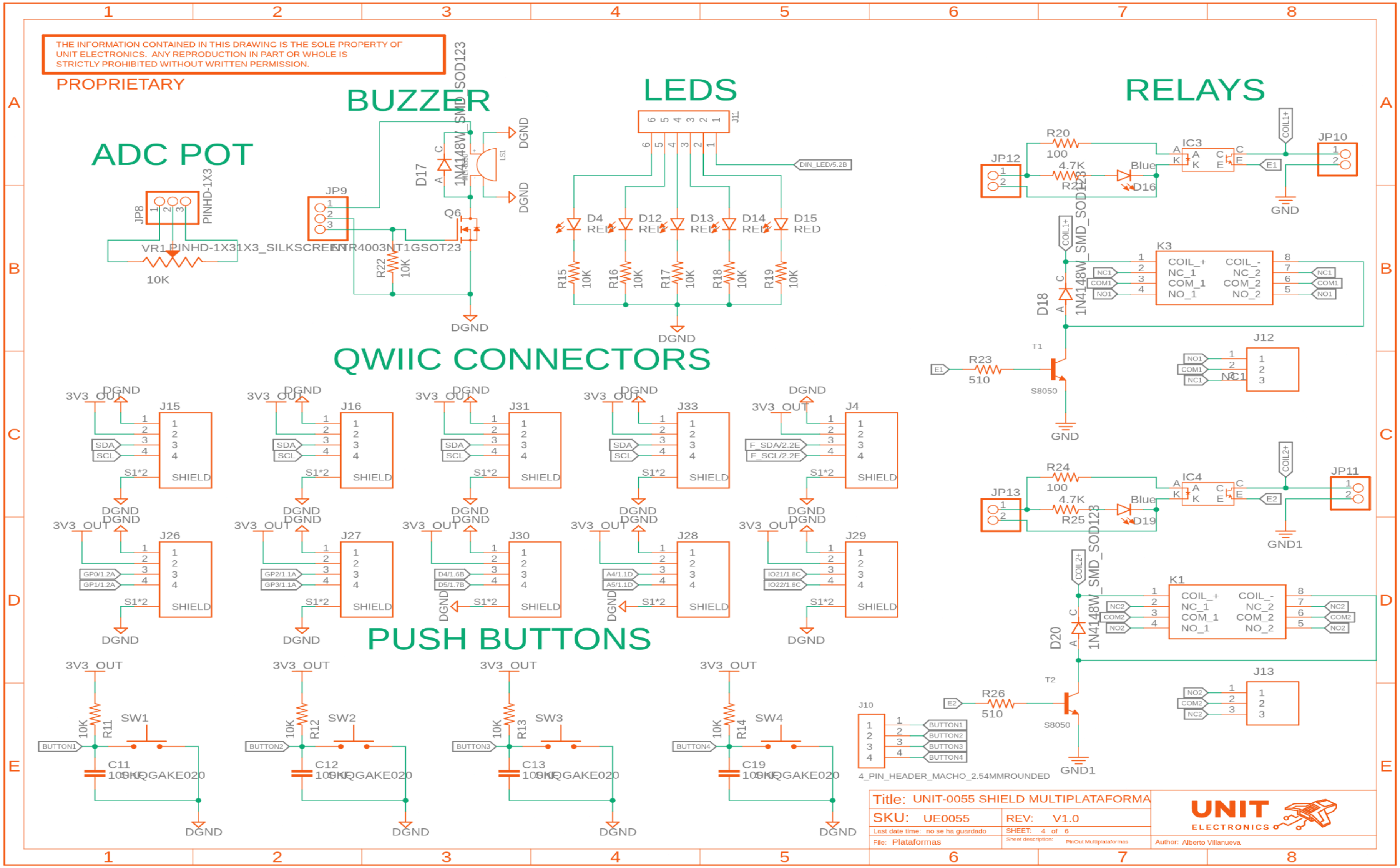
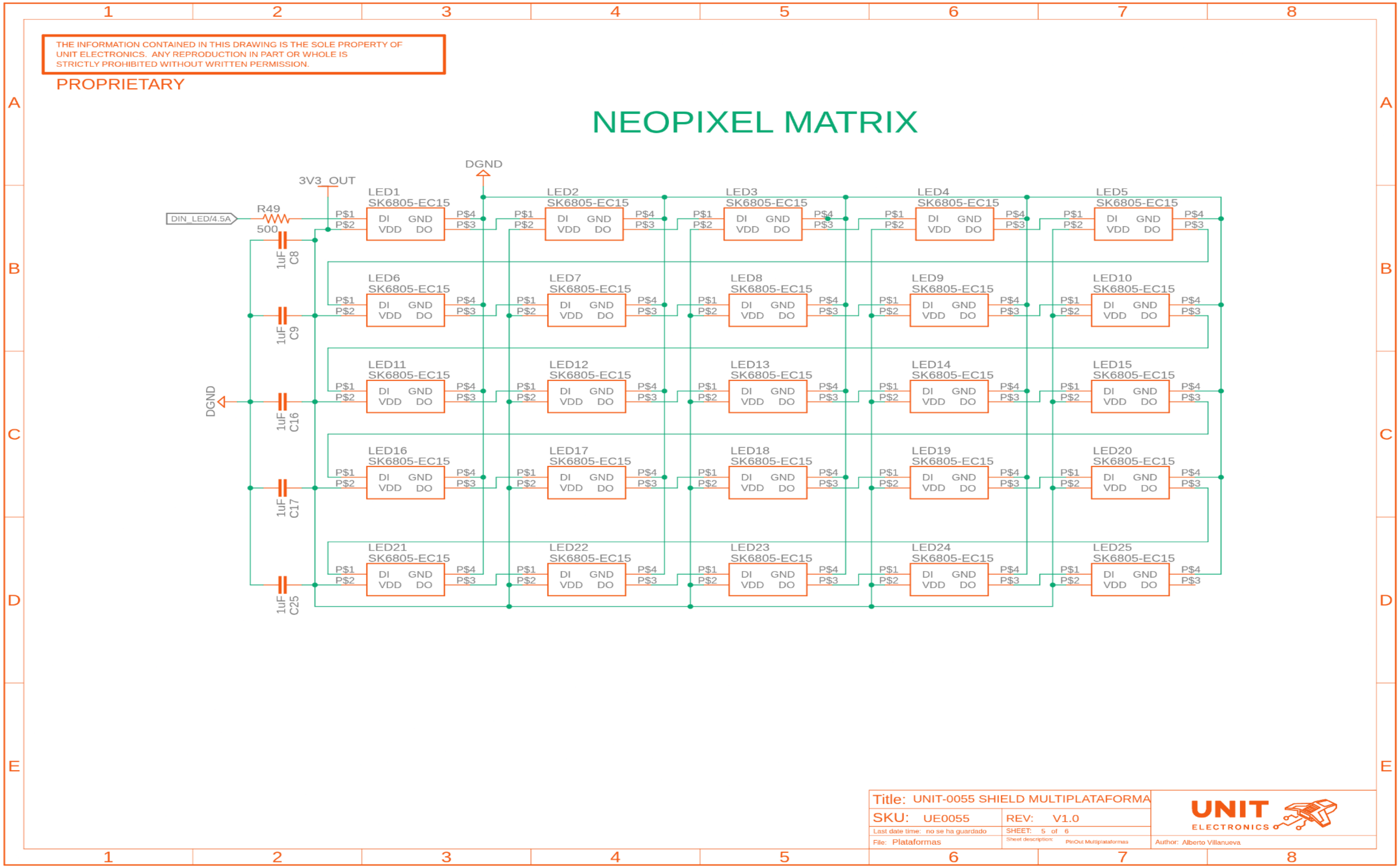
PROPRIETARY
USB C
OUTPUT SELECTION
VOLTAGE REGULATORS
DECOY
POWER ON
OUT
U1
TPS54302DDCT
U2
TPS54302DDCT
IC2
HUSB238
Q2
AONR21307
DIP_SWITCH_5_DIGITS_SPST_1.27P
Resistor VSET
0 5V
6.04K 9V
10K 12V
14K 15V
17.8K 18V
Open 20V
JP1
PINHD-1X31X3_SILKSCREEN
BORNERA-KF301-2P
Title: UNIT-0055 SHIELD MULTIPLATAFORMA
SKU: UE0055
REV: V1.0
File: Plataformas
Author: Alberto Villanueva


THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF UNIT ELECTRONICS. ANY REPRODUCTION IN PART OR WHOLE IS STRICTLY PROHIBITED WITHOUT WRITTEN PERMISSION.
PROPRIETARY
ADC POT
BUZZER
LEDS
RELAYS
QWIIC CONNECTORS
PUSH BUTTONS
Title: UNIT-0055 SHIELD MULTIPLATAFORMA
SKU: UE0055
REV: V1.0
Last date time: no se ha guardado
SHEET: 4 of 6
File: Plataformas
Author: Alberto Villanueva
UNIT ELECTRONICS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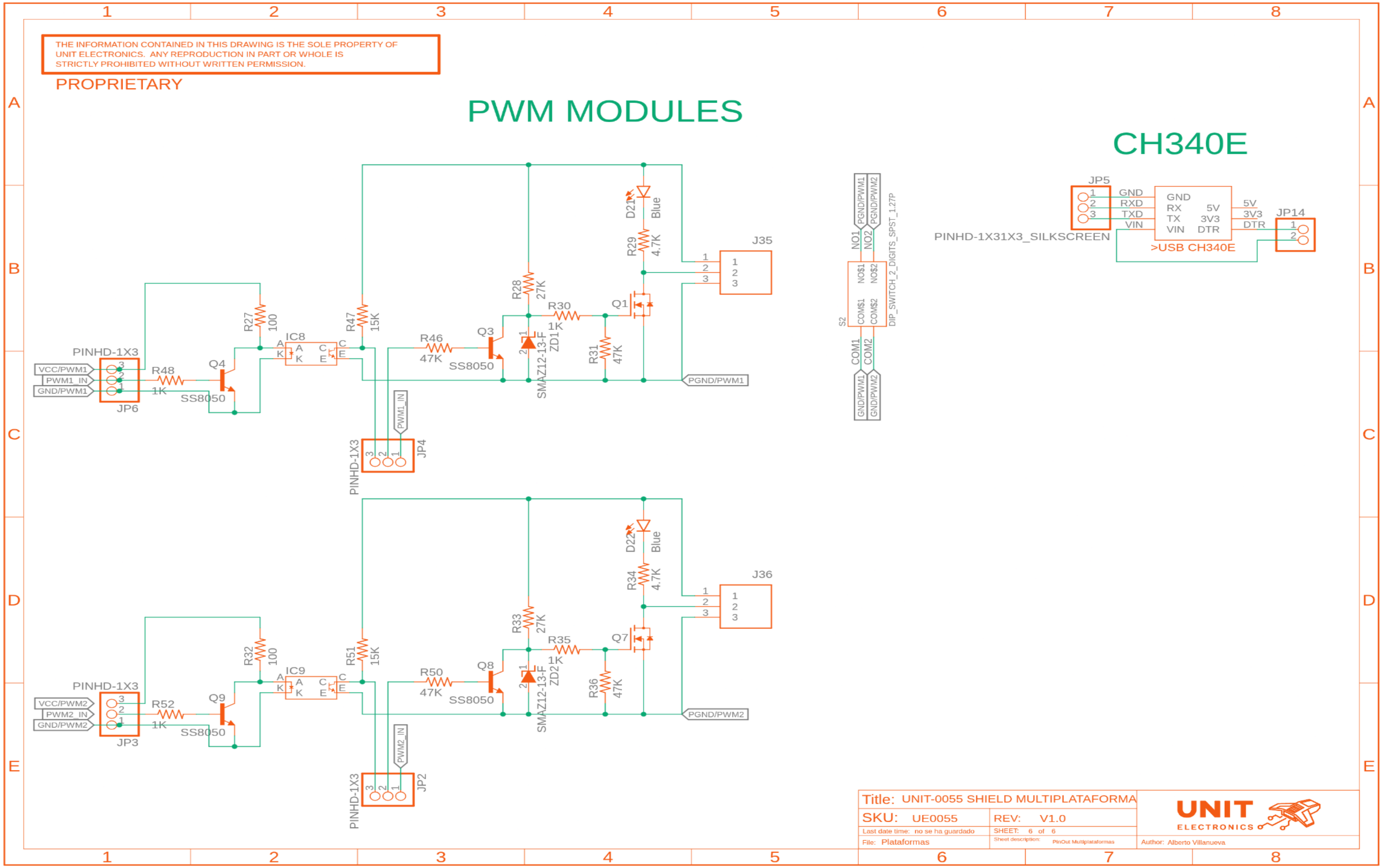
PROPRIETARY
NEOPIXEL MATRIX
DGND
3V3_OUT
DIN_LED/4.5A
R49
500
1uF C8
1uF C9
1uF C16
1uF C17
1uF C25
LED1 SK6805-EC15
LED2 SK6805-EC15
LED3 SK6805-EC15
LED4 SK6805-EC15
LED5 SK6805-EC15
LED6 SK6805-EC15
LED7 SK6805-EC15
LED8 SK6805-EC15
LED9 SK6805-EC15
LED10 SK6805-EC15
LED11 SK6805-EC15
LED12 SK6805-EC15
LED13 SK6805-EC15
LED14 SK6805-EC15
LED15 SK6805-EC15
LED16 SK6805-EC15
LED17 SK6805-EC15
LED18 SK6805-EC15
LED19 SK6805-EC15
LED20 SK6805-EC15
LED21 SK6805-EC15
LED22 SK6805-EC15
LED23 SK6805-EC15
LED24 SK6805-EC15
LED25 SK6805-EC15
P$1 P$2 DI VDD GND DO P$4 P$3
Title: UNIT-0055 SHIELD MULTIPLATAFORMA
SKU: UE0055
REV: V1.0
Last date time: no se ha guardado
SHEET: 5 of 6
Sheet description: PinOut Multiplataformas
File: Plataformas
Author: Alberto Villanueva
UNIT ELECTRONICS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PROPRIETARY
PWM MODULES
CH340E
JP5
GND
RXD
TXD
VIN
GND
RX
TX
VIN
5V
3V3
DTR
JP14
PINHD-1X31X3_SILKSCREEN
>USB CH340E
J35
J36
D21
D22
Blue
R29 4.7K
R34 4.7K
R28 27K
R33 27K
R30 1K
R35 1K
Q1
Q7
R31 47K
R36 47K
Q3
Q8
SS8050
R46 47K
R50 47K
ZD1
ZD2
SMAZ12-13-F
R47 15K
R51 15K
IC8
IC9
R27 100
R32 100
Q4
Q9
R48 1K
R52 1K
PINHD-1X3
JP6
JP3
JP4
JP2
VCC/PWM1
PWM1_IN
GND/PWM1
VCC/PWM2
PWM2_IN
GND/PWM2
PGND/PWM1
PGND/PWM2
PWM1_IN
PWM2_IN
S2
DIP_SWITCH_2_DIGITS_SPST_1.27P
NO1
NO2
COM1
COM2
Title: UNIT-0055 SHIELD MULTIPLATAFORMA
SKU: UE0055
REV: V1.0
SHEET: 6 of 6
Last date time: no se ha guardado
File: Plataformas
Author: Alberto Villanueva